HEXLEY CHEMICALS

Permit Rules

Document Rules 01 | Issue 5 | Owner: Permit Office | Review cycle: every twelve months

1. Purpose

These rules say how a permit request and the records of past permits are read: how each of the three channels writes each answer and what it is read as, which revision of a request is the request, how each requirement of a permit standard is judged Met, Not met or Not documented, and how the documents attached to a request are read against it.

They hold no requirement of their own. Every requirement and its figures are written in the permit standard of the request's own permit type, and the worked example of section 13 uses figures that belong to no permit type.

2. The three channels

The term contractors resident on site request permits through the site portal, the specialist contractors through the contractor app, and the small local contractors on paper forms that the permit office scans: scanned paper. Each channel writes the same request its own way. An answer is read the same whichever channel wrote it, and a spelling these rules do not list is refused, naming it, never guessed.

The plant area is one of the works' named areas and the work activity one of its permit type's jobs, each written by its name in every channel. The permits live beside a request are counted from the live permit register; the performing authority gives everything else.

3. References

Every request takes its number from the permit system. The site portal writes PTW and seven digits, as in PTW0050118; the contractor app writes PTW- and the number, as in PTW-50118; and scanned paper writes the bare number, 50118 in the same case. Each is read as PTW and the number in seven digits, so all three are PTW0050118.

Contractors are written CTR- and four digits, documents DOC- and six digits, and statement sentences SEN- and seven digits, the same in every channel.

4. Gas readings

The flammable gas reading is read as a whole percent of the lower explosive limit, a reading with a decimal part rounded half up once: the site portal writes the number alone, as in 3; the contractor app writes 3%LEL; and scanned paper writes 3 % LEL.

The oxygen reading is read as percent, rounded half up once to one decimal: 20.9 from the site portal, 20.9% from the contractor app and 20.9 % O2 from scanned paper.

Hydrogen sulphide and carbon monoxide are read as ppm, rounded half up once to one decimal. The site portal writes the number alone, as in 2.5, and the contractor app writes 2.5 ppm. Scanned paper writes milligrams per cubic metre, as in 3.55 mg/m3, converted exactly - hydrogen sulphide divided by 1.42 and carbon monoxide by 1.17 - and then rounded half up once to one decimal: 3.55 mg/m3 of hydrogen sulphide is 2.5 ppm and 7.2 mg/m3 is 5.1 ppm; 4.68 mg/m3 of carbon monoxide is 4.0 ppm and 23.5 mg/m3 is 20.1 ppm.

5. Lengths, heights and wind

The clearance of combustible materials, the working height and the depth of a dig are read as metres to one decimal: the site portal writes the number alone, as in 10; the contractor app writes 10 m; and scanned paper writes feet, as in 33 ft. A length in feet is multiplied by 0.3048 exactly and rounded half up once to one decimal: 33 ft is 10.1 m, 32 ft is 9.8 m, 3.9 ft is 1.2 m and 4.2 ft is 1.3 m.

The wind speed is read as metres per second to one decimal: 6.2 from the site portal, 6.2 m/s from the contractor app, and miles per hour from scanned paper, as in 14 mph. A wind in miles per hour is multiplied by 0.44704 exactly and rounded half up once to one decimal: 14 mph is 6.3 m/s, 26.8 mph is 12.0 m/s and 28 mph is 12.5 m/s.

6. Voltage

The system voltage is read as whole volts. The site portal writes volts alone, as in 415; the contractor app writes volts below 1,000 volts, as in 415V, and kilovolts from 1,000 volts, as in 11kV; and scanned paper writes kilovolts, as in 0.415 kV. A voltage in kilovolts is multiplied by 1,000 exactly and rounded half up once to whole volts: 0.415 kV is 415 volts, 3.3kV is 3,300 volts and 1kV is 1,000 volts.

7. Times and counts

The fire watch is read as whole minutes: the site portal writes the number alone, as in 60; the contractor app writes 60 min; and scanned paper writes hours, as in 1 hr or 1.5 hrs, multiplied by 60 and rounded half up once to whole minutes, so 1.5 hrs is 90 minutes and 0.75 hrs is 45 minutes.

The planned and actual work hours are read as hours to one decimal: the site portal writes the number alone, as in 6.5; the contractor app writes minutes, as in 390 min, divided by 60 and rounded half up once to one decimal; and scanned paper writes 6.5 hrs, or 1 hr for a single hour.

The size of the work party, a revision number, the permits live beside a request and the number of documents attached are whole numbers.

8. Named answers

The permit type is written in full by the site portal - Hot work, Confined space entry, Electrical isolation, Work at height or Excavation - as a permit's name by the contractor app - Hot Work Permit, Confined Space Permit, Electrical Isolation Permit, Work at Height Permit or Excavation Permit - and as a code on scanned paper - HW, CSE, EI, WAH or EXC. Each is read as the permit type in full. The linked permit type is written the same way, and no linked permit is written No linked permit, No link or NIL.

The hazardous area classification is Zone 1, Zone 2 or Non-hazardous in the site portal; Z1, Z2 or Safe area in the contractor app; and ZONE-1, ZONE-2 or NON-HAZ on scanned paper.

The process unit status is Operating, Shut down or No process plant in the site portal; Running, Off-line or No plant in the contractor app; and OPER, S/D or NO PLANT on scanned paper.

The work location is Indoors or Outdoors in the site portal, Inside or Outside in the contractor app, and IN or OUT on scanned paper.

The access method is Scaffold, Mobile elevating work platform, Ladder, Fixed platform or Rope access in the site portal; Scaffold, MEWP, Ladder, Fixed platform or Rope access in the contractor app; and SCAF, MEWP, LADDER, PLATFORM or ROPE on scanned paper.

Yes is written Yes by the site portal, true by the contractor app and Y on scanned paper, and no is No, false and N; yes, y, true and 1 are also read as Yes, and no, n, false and 0 as No.

The records of past permits add the answers given after the decision. The review outcome is Issued as requested, Issued after rework or Issued after SIMOPS review in the site portal; Issued, Issued - rework or Issued - SIMOPS in the contractor app; and ISS, ISS-RW or ISS-SIM on scanned paper. The suspension reason is Gas alarm, Process upset, Site alarm, Conflicting work, Unsafe condition or High wind in the site portal; Gas alarm, Process upset, Site alarm, Conflict, Unsafe or Wind in the contractor app; and GAS, PROC, ALARM, SIMOPS, UNSAFE or WIND on scanned paper.

A job log's completion code is Completed, Completed after extension or Completed after suspension in the site portal; Done, Done - extended or Done - suspended in the contractor app; and COMP, COMP-EXT or COMP-SUS on scanned paper.

The permit office's labels on past statement sentences are written several ways, each read as one of the nine sentence types: Isolation as "ISOLATION", "Isolations" or "isolation"; Gas testing as "Gas Testing", "Gas test" or "gas-testing"; Fire precautions as "Fire Precautions", "Fire" or "fire-precautions"; Rescue as "RESCUE", "Emergency rescue" or "rescue"; Access as "ACCESS", "Access and egress" or "access"; Personal protective equipment as "PPE", "Personal Protective Equipment" or "personal-protective-equipment"; Simultaneous operations as "SIMOPS", "Simultaneous Operations" or "simultaneous-operations"; Housekeeping and waste as "Housekeeping & waste", "Housekeeping" or "housekeeping-and-waste"; Background as "Context", "General" or "background". Each type is also read when written as its own name.

A document's type is Method statement, Task risk assessment, Gas test record or Rescue plan, and a permit's status in the live permit register is Live or Suspended.

Every named answer is read whatever its capitals and with any spaces around it trimmed, and so are the plant area and the work activity.

9. Missing answers and impossible readings

An answer left blank is written N/A or not recorded by the site portal, left empty or written null by the contractor app, and written - or illegible on scanned paper; NA is also read as blank. A question the request does not ask is written N/A by the site portal, left empty by the contractor app and written - on scanned paper, and is blank too.

An impossible reading is made blank, never corrected: a negative gas reading, a flammable gas or oxygen reading above 100 percent, a clearance, working height or depth of nought or less, a negative wind speed, a fire watch, system voltage or planned work hours of nought or less, and a work party of nobody.

A record of a past permit whose permit suspended answer is blank, or a job log whose actual work hours are blank or impossible, cannot be learned from and is left out. So is a past statement sentence of fewer than three words, or one whose type is blank.

10. Revisions and repeats

A request revised before its review is sent again with a higher revision number. The highest revision is the request, whatever an earlier revision said and wherever the rows sit in the file; the earlier revisions are never screened. A request returned to its requester and sent again with its rework takes the next revision number in the same way. A row repeated exactly is one row, in the live files and in the records alike.

11. Requirement results

Each requirement of the permit type's own standard, and of its linked type's, is Met, Not met or Not documented:

- A yes-or-no requirement is Met on Yes, Not met on No and Not documented when the answer is blank.
- A maximum is met at its figure and a minimum likewise: a reading of exactly the figure is within it, and a band is met at either of its figures. Every reading is compared in the unit its standard states, converted where it was written in another, and rounded half up once to the precision these rules read it at - a whole percent of the lower explosive limit, whole minutes and whole volts, and a tenth of a percent of oxygen, of a ppm, of a metre, of a metre per second and of an hour - whether it was converted or not. Every figure is compared as an exact decimal.
- Circuit earths are a follow-on question decided by the system voltage: Met at or below the standard's voltage whatever else the request says; above it Met with circuit earths applied, Not met without them and Not documented when that answer is blank; and Not documented, asking for the voltage alone, when the voltage is blank.
- Scaffold inspection is decided by the access method: Met for any access but a scaffold; from a scaffold Met when it is inspected, Not met when it is not and Not documented when that answer is blank; and Not documented, asking for the access method alone, when the access method is blank.
- Wind speed is decided by the work location: Met indoors; outdoors Met at or below the standard's figure, Not met above it and Not documented when the wind is blank; and Not documented, asking for the work location alone, when the location is blank.
- Trench support is decided by the depth: Met at a depth of at most the standard's figure; deeper Met when the sides are supported, Not met when they are not and Not documented when that answer is blank; and Not documented, asking for the depth alone, when the depth is blank.

A requirement that is Not documented names the answers it needs that were left blank, and only those. A request with a linked permit type is screened under its own standard first and then under its linked type's, and each requirement keeps the standard it came from. Then the assessment follows, as the Control of Work Policy sets out: any requirement Not met or Not documented makes it RETURN TO REQUESTER; otherwise any SIMOPS trigger makes it REFER FOR SIMOPS REVIEW; otherwise it is ISSUE PERMIT.

12. Statements against the request

Every document is read as plain words, without markup, character entities or no-break spaces, and split into its sentences in order: a sentence ends at a full stop, a question mark or an exclamation mark followed by a space and a letter or a digit, whatever its case. Each sentence is given one of nine types: isolation, gas testing, fire precautions, rescue, access, personal protective equipment, simultaneous operations, housekeeping and waste, or background. Each type bears on the requirements its statement guide names; personal protective equipment, simultaneous operations, housekeeping and waste, and background bear on none.

A sentence typed as gas testing is searched for readings, each written in one of four forms, in any capitals: a number followed by % LEL, a number followed by % oxygen, a number followed by ppm hydrogen sulphide, or a number followed by ppm carbon monoxide, with no space before the percent sign and one space between the other parts. Each reading is compared as it is written, without rounding, as an exact decimal, with the request's reading of the same gas. Only gas testing sentences are searched, and a sentence of any other type is never flagged.

A sentence with no reading is not flagged. One with a reading is flagged DESCRIBES BLANK ANSWER where the request's reading of a gas it reads is blank - left blank, or not asked of its permit types - and otherwise DISAGREES WITH FORM when any of its readings differs from the request's. Against a request of 0 and 20.9, "The pre-job gas test read 0% LEL and 20.9% oxygen." is not flagged; against a request of 2 and 20.9, it disagrees with the form. A flag never changes an answer: the request is what the performing authority signed, and the disagreement is for the Area Authority to settle.

13. A worked example

This example uses a standard that belongs to no permit type, made up so that every reading is seen at work at once, outside any real standard: a flammable gas reading of at most 8 percent of the lower explosive limit, an oxygen reading of at least 19.0 percent and at most 23.0 percent, a hydrogen sulphide reading of at most 3 ppm, and, outdoors, a wind of at most 10 metres per second.

A hot work request comes on scanned paper, written HW, for work outdoors, written OUT, on a process unit written OPER. Its flammable gas reading, 8 % LEL, is exactly 8, at the limit: Met. Its oxygen, 18.9 % O2, is under 19.0: Not met. Its hydrogen sulphide, written 4.3 mg/m3, is 4.3 divided by 1.42, or 3.028, which rounds half up once to 3.0 ppm, at the limit: Met. Its wind, which the made-up standard asks about only to show the reading, is written illegible, which is blank: Not documented, asking for the wind speed and only the wind speed.

That is Met 2, Not met 1 and Not documented 1, so the assessment is RETURN TO REQUESTER. The process unit is operating, so the live process unit trigger applies too - but the requirements come first.